FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

INTERMEDIATE (12TH CLASS) EXAMINATION, 2025

MATHEMATICS

Past Paper 2025 • Paper Code: 2HA-I 25008 — 40081 (Version B) / 42081 (Version D)

Retyped faithfully from the original board paper — same questions, no changes.

VERSION B

MATHEMATICS HSSC-II — SECTION – A (Objective Type)

Time Allowed: 25 Minutes | **Maximum Marks:** 20 | **Paper Code:** 40081 (Version B)

Note: Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed. Do not use lead pencil.

حصہ اول لازمی ہے۔ اس کے جوابات اسی صفحہ پر دے کر ناظم مرکز کے حوالے کریں۔ کاٹ کر دوبارہ لکھنے کی اجازت نہیں ہے۔ لیڈ پنسل کا استعمال ممنوع ہے۔

Note: Fill the relevant bubble against each question according to curriculum:

ہر سوال کے سامنے دیے گئے کریکولم کے مطابق درست دائرہ کو پر کریں۔

Q1. lim(x→a) ((1+2x)^2 −1)/x =

(A) 1 (B) 2

(C) 4 (D) 6

Q2. lim(x→∞) (1+2/x)^(x/2) =

(A) 0 (B) 1

(C) e (D) 1/e

Q3. d/dx (e^x + 2e^(−2x)) =

(A) e^x − 4e^(−2x) (B) e^x + 4e^(−2x)

(C) e^x − e^(−2x)/2 (D) e^x − 2e^(−2x)

Q4. If $dy/dx = 3/2\ x^{(1/2)} - 2x$, then $d^2y/dx^2 =$

(A) $3/2\ x^{(1/2)} - 2$
(B) $3/4\ x^{(-1/2)} - 2$
(C) $1/4\ x^{(1/2)} - 2$
(D) $4/3\ x^{(-1/2)} - 2$

Q5. If $y = 1/8 \sin 2x$, then the value of d^4y/dx^4 is:

(A) $2 \sin 2x$
(B) $-2 \sin 2x$
(C) $2 \cos 2x$
(D) $-2 \cos 2x$

Q6. If $\vec{r}(t) = (1+t^2)\hat{i} + \sin t\,\hat{j} + \cos 2t\,\hat{k}$, then $\lim(t\to 0)\ \vec{r}(t) =$

(A) $\hat{i} + \hat{j} + \hat{k}$
(B) $\hat{i} + \hat{k}$
(C) $\hat{i} + \hat{j} - \hat{k}$
(D) $\hat{i} + \hat{j}$

Q7. If $\vec{r}(t) = t^2\hat{i} - 2t\hat{j}$, then $\vec{v}(t) =$

(A) $2t\hat{i} - \hat{j}$
(B) $2t\hat{i} - 2\hat{j}$
(C) $t\hat{i} - 2\hat{j}$
(D) $t\hat{i} - \hat{j}$

Q8. $\int \cos x / \sin x\ dx =$

(A) $\ln|\sin x| + c$
(B) $\ln|\cos x| + c$
(C) $-\ln|\sin x| + c$
(D) $-\ln|\cos x| + c$

Q9. If $2\int_0^2 x\ dx = 3k - 2$, then the value of 'k' is:

(A) 2
(B) 3
(C) 4
(D) 6

Q10. If A(5,−5), B(−2,0) and C(0,2) are vertices of a triangle then its centroid is:

(A) (1,1)
(B) (−1,−1)
(C) (−1,1)
(D) (1,−1)

Q11. If $4x - 2y = 3$, then the slope of the line is:

(A) −3
(B) −2
(C) 2
(D) 3

Q12. The equation of circle with centre (−1,−1) and radius 2 is:

(A) $(x-1)^2 + (y-1)^2 = 2$
(B) $(x-1)^2 + (y-1)^2 = 4$
(C) $(x+1)^2 + (y+1)^2 = 2$
(D) $(x+1)^2 + (y+1)^2 = 4$

Q13. If $x^2 + y^2 + 4x + 4y = 0$ is an equation of circle then its radius is:

(A) 2
(B) $2\sqrt{2}$
(C) 3
(D) $3\sqrt{2}$

Q14. If $y^2 = 8x$ is an equation of parabola, then its focus is:

(A) (0,4) (B) (0,2)

(C) (2,0) (D) (4,0)

Q15. If $d^2y/dx^2 - (dy/dx)^2 + y = 3x$ is a differential equation, then its order and degree respectively are:

(A) 1,2 (B) 2,1

(C) 2,2 (D) 2,3

Q16. If $1/x\ dy/dx = 2$, then solution y(x) of differential equation is:

(A) $x^2 + c$ (B) $x^2/2 + c$

(C) $2x^2 + c$ (D) $-x^2 + c$

Q17. If $f(x,y) = x^2 + 3y$, then $\partial f/\partial x\ (1,1) =$

(A) 0 (B) 2

(C) 3 (D) 5

Q18. If f(x,y) = e^(x+y), then $\partial f/\partial y\ (1,0)$ is:

(A) 0 (B) 1

(C) e (D) 1/e

Q19. If $f(x) = x^3 - 2x + 1$ and $x_0 = 0$, then $f'(x_0)$ is:

(A) −2 (B) −1

(C) 1 (D) 2

Q20. The value of Δx to approximate $\int_0^2 (x^2 + 1)\ dx$ for (n = 4 sub intervals) by 'Trapezoidal Rule' is:

(A) 0 (B) 0.25

(C) 0.5 (D) 1

> Note: $\vec{v}(t) = d/dt\ (\vec{r}(t))$

MATHEMATICS HSSC-II — SECTIONS B & C (Subjective Type)

Time Allowed: 2:35 Hours | **Maximum Marks:** 80 | **Paper Code:** 2HA-I 25008 (B)

> Note: SECTION – B (Marks 48) — Q.2 Solve the following:

(i) [04] Evaluate: $\lim(\theta\to0)\ (1-\cos\theta)/\sin^2\theta$ — OR (1+3) — If $f(x,y) = 3x^3 + 7x^2y + xy^2 + 5y^3$, then find: a. Degree of homogeneous function b. Verify Euler's theorem for f(x,y)

(ii) [04] Find the value of 'k' if f(x) is continuous at x=1: f(x) = { $(3x^2-3)/(x-1)$, $x\neq1$; $2(k-1)x$, x=1 } — OR (04) — If line $x-2y+l=0$ is tangent to the ellipse $x^2/16 + y^2/9 = 1$, then find: a. Value of l b. Equation of tangent c. Length of latus rectum

(iii) [04] If $y = 3\theta^3 - \theta^{(-3/2)}$ and $u = 2\theta^2 - \theta^{(-1/2)}$ then find dy/du — OR (04) — Compute two iterations up to 4 decimal places of $f(x) = x^3 - 5x + 1$ by using Newton Raphson's method with initial start $x_0 = 0.5$

(iv) [04] If $\vec{r}(t) = t^3\hat{i} + e^{(-2t)}\hat{j} + \sin 2t\,\hat{k}$ is position vector of a moving particle at time (t), then find: a. Velocity vector $\vec{v}(t)$ b. Acceleration vector $\vec{a}(t)$ — OR (3+1) — Find the equation of line passing through the point of intersection of lines x+y=3 and x−y=1 and parallel to line x−3y=2 and write it into slope intercept form.

(v) [04] Evaluate: $\int(3t+1)(t-1)\,dt$ — OR (04) — Find the equation of the circle passing through the points (4,0), (0,5) and (0,0)

(vi) [04] Find the solution of differential equation dy/dx = 1 + e^x y + y + e^x — OR (04) — Find the area under the graph of $f(x) = 3x^2 - 2x$ over the interval [1,3]

(vii) [04] Find the extreme values of $f(x) = 1/3\ x^3 - x^2 - 3x + 6$ — OR (04) — If $3x^2 - 2xy - 8y^2 = 0$ is 2nd degree homogeneous equation then find: a. Pair of straight lines b. Angle between these lines

(viii) [04] Find the area of a triangle if its vertices are A(5,4), B(2,1) and C(7,3) — OR (04) — Evaluate $\int x^2 \cos x\,dx$ using integration by parts method

(ix) [04] Find the equation of tangent and normal to parabola $x^2 = 8y - 8$ at x = 4 — OR (04) — Find dy/dx, if y = (2x+3)^(3/2) by first principle

(x) [04] Evaluate $\int e^x\,dx / (e^{(2x)} + 1)$ by substitution method — OR (04) — If $x^2/9 - y^2/4 = 1$ is an equation of hyperbola then find: a. Centre b. Vertices c. Foci d. Equation of asymptotes

(xi) [04] If $f(x,y) = \sqrt{(x^3 + y^3)}$, then find: a. $\partial f/\partial x$ b. $\partial f/\partial y$ — OR (04) — If f(x) = 2x−3 and g(x) = 3x+2, and h(x) = f(g(x)) then find (i) h(x) (ii) $h^{-1}(x)$ (iii) value of x if $h^{-1}(x) = 2$

(xii) [04] Compute four iterations of the function $f(x) = 2x^3 - 5$ in the interval [0,2] by using Bisection method — OR (04) — Find dy/dx, if $x^3 + 3x^2y^2 + y^3 = 0$

> Note: SECTION – C (Marks 32) — Note: Solve the following Questions. (4 x 8 = 32)

Q3. (Marks 08) Q.3 The growth of bacteria is given by the function of time (t) (in hours) f(t) = 500e^(0.25t), then find: a. Initial number of Bacteria at t=0 b. Number of Bacteria after t=4 hours c. Number of Bacteria after t=8 hours d. The time if number of Bacteria is $500e^3$. e. Is f(t) continuous at t=4 hours and t=8 hours? — OR (08) — If $f(x) = 1/8\ x^3 + 1/4\ x^2 + 2/x^2 + 1/2$, then find: a. Equation of tangent at x=2 in ax+by=c form b. Equation of normal at x=2 c. Test continuity of f(x) at x=0

Q4. (Marks 08) Q.4 In the given ΔABC find equation of: a. Median $C\bar{M}$ b. Altitude $C\bar{D}$ c. Right bisector of $A\bar{B}$ at mid-point, M — [Graph: A(1,2), B(7,2), C(3,6)] — OR (2+2+2+1+1) — In an elliptic ground if a = 2b and b = 4 then find: a. Vertices, co-vertices and centre b. Foci c. Eccentricity d. Equation in standard form with center

at origin e. Draw the rough diagram of ellipse using vertices and co-vertices only. (Graph paper is not required)

Q5. (Marks 08) Q.5 Evaluate: $\int (2x-1)/(x^3 - x^2 - 2x)\, dx$ — OR (08) — Solve the homogeneous differential equation $2xy\, dy - (x^2 + 3y^2)\, dx = 0$

Q6. (Marks 08) Q.6 Evaluate: $\int dx / (4x^3 - x)$ — OR (5+2+1) — a. Approximate $\int_0^2 (3x^2 - 2x)\, dx$ for (n=2 sub intervals) by using Simpson's Rule b. Find actual value of definite integral c. Calculate difference of both values

VERSION D

MATHEMATICS HSSC-II — SECTION – A (Objective Type)

Time Allowed: 25 Minutes | **Maximum Marks:** 20 | **Paper Code:** 42081 (Version D)

Note: Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed. Do not use lead pencil.

حصہ اول لازمی ہے۔ اس کے جوابات اسی صفحہ پر دے کر ناظم مرکز کے حوالے کریں۔ کاٹ کر دوبارہ لکھنے کی اجازت نہیں ہے۔ لیڈ پنسل کا استعمال ممنوع ہے۔

Note: Fill the relevant bubble against each question according to curriculum:

ہر سوال کے سامنے دیے گئے کریکولم کے مطابق درست دائرہ کو پر کریں۔

Q1. $\lim(x\to0)\ ((1+3x)^2 - 1)/x =$

(A) 1 (B) 2

(C) 3 (D) 6

Q2. $\lim(x\to\infty)\ (1+3/x)^{(x/3)} =$

(A) 0 (B) 1

(C) e (D) 1/e

Q3. $d/dx\ (e^x + 3e^{(-3x)}) =$

(A) $e^x - 9e^{(-3x)}$ (B) $e^x + 9e^{(-3x)}$

(C) $e^x - 6e^{(-3x)}$ (D) $e^x + 6e^{(-3x)}$

Q4. If $dy/dx = 2x + 5/2\ x^{(3/2)}$, then $d^2y/dx^2 =$

(A) $2 + 5/2\ x^{(1/2)}$ (B) $2 + 15/4\ x^{(1/2)}$

(C) $2 - 15/4\ x^{(1/2)}$ (D) $2 + 5/2\ x^{(3/2)}$

Q5. If $y = 1/8 \cos 2x$, then the value of d^4y/dx^4 is:

(A) $2 \cos 2x$ (B) $-2 \cos 2x$

(C) $2 \sin 2x$ (D) $-2 \sin 2x$

Q6. If $\vec{r}(t) = (t^2-1)\hat{i} + \sin t\,\hat{j} + \cos 2t\,\hat{k}$, then $\lim(t\to 0)\ \vec{r}(t)$ is =

(A) $\hat{i} + \hat{k}$ (B) $-\hat{i} + \hat{k}$

(C) $-\hat{i} + \hat{j} + \hat{k}$ (D) $-\hat{i} - \hat{j} + \hat{k}$

Q7. If $\vec{r}(t) = t^2\hat{i} + 2t\hat{j}$, then $\vec{v}(t) =$ is:

(A) $2t\hat{i} + \hat{j}$ (B) $2t\hat{i} + 2\hat{j}$

(C) $t\hat{i} + 2\hat{j}$ (D) $t\hat{i} + \hat{j}$

Q8. $\int \sin t / \cos t\, dt$ is equal to:

(A) $-\ln|\cos t| + c$ (B) $\ln|\cos t| + c$

(C) $\ln|\sin t| + c$ (D) $-\ln|\sin t| + c$

Q9. If $2\int_0^3 x\, dx = 2k - 3$, then the value of 'k' is:

(A) 2 (B) 3

(C) 4 (D) 6

Q10. If A(−5,5), B(2,0) and C(0,−2) are vertices of a triangle then its centroid is:

(A) (1,1) (B) (−1,−1)

(C) (1,−1) (D) (−1,1)

Q11. If $6x - 3y = 2$, then the slope of the line is:

(A) −3 (B) −2

(C) 2 (D) 3

Q12. The equation of circle with centre (1,1) and radius 2 is:

(A) $(x+1)^2 + (y+1)^2 = 2$ (B) $(x+1)^2 + (y+1)^2 = 4$

(C) $(x-1)^2 + (y-1)^2 = 2$ (D) $(x-1)^2 + (y-1)^2 = 4$

Q13. If $x^2 + y^2 + 6x + 6y = 0$ is an equation of circle then its radius is:

(A) 2 (B) $2\sqrt{2}$

(C) 3 (D) $3\sqrt{2}$

Q14. If $y^2 = -8x$ is an equation of parabola, then its focus is:

(A) (0,−4) (B) (0,−2)

(C) (−2,0) (D) (−4,0)

Q15. If $d^3y/dx^3 - d^2y/dx^2 + (dy/dx)^2 = x^4$ is a differential equation, then its order and degree respectively are:

(A) 3,1 (B) 1,3

(C) 2,3 (D) 3,2

Q16. If $1/x^2\ dy/dx = 3$ then solution y(x) of differential equation is:

(A) $x^3 + c$ (B) $x^3/3 + c$

(C) $3x^3 + c$ (D) $-x^3 + c$

Q17. If $f(x,y) = 3x + y^2$, then $\partial f/\partial y\ (1,1)$ is:

(A) 0 (B) 2

(C) 3 (D) 5

Q18. If f(x,y) = e^(x−y), then $\partial f/\partial x\ (0,1)$ is:

(A) 0 (B) 1

(C) e (D) 1/e

Q19. If $f(x) = x^3 - 2x + 3$ and $x_0 = 0$, then $f'(x_0)$ is:

(A) −2 (B) −1

(C) 1 (D) 2

Q20. The value of Δx to approximate $\int_0^2 (x^2 - 1)\ dx$ for (n = 4 sub intervals) by 'Trapezoidal Rule' is:

(A) 0 (B) 0.25

(C) 0.5 (D) 1

MATHEMATICS HSSC-II — SECTIONS B & C (Subjective Type)

Time Allowed: 2:35 Hours | **Maximum Marks:** 80 | **Paper Code:** 2HA-I 25008 (D)

Note: SECTION – B (Marks 48) — Q.2 Solve the following:

(i) [04] Evaluate: $\lim(\theta\to0)\ \sin^2\theta/(1-\cos\theta)$ — OR (1+3) — If $f(x,y) = 2x^3 - 5xy^2 + 3x^2y - 4y^3$ then: a. Find degree of homogeneous function b. Verify Euler's theorem for f(x,y)

(ii) [04] Find the value of 'k' if $f(x) = \{ (2x^2-2)/(x-1),\ x\neq1;\ 3(k-1)x,\ x=1 \}$ is continuous at x=1 — OR (04) — If line x−3y+l=0 is tangent to the ellipse $x^2/25 + y^2/16 = 1$, then find: a. Value of l b. Equation of tangent. c. Length of latus rectum

(iii) [04] If $y = 1/3\ \theta^3 - \theta^{(-1/2)}$ and $z = 1/2\ \theta^2 - \theta^{(-3/2)}$ then find dy/dz. — OR (04) — Compute two iterations up to 4 decimal places of $f(x) = 2x^3 - 5x + 1$ by using Newton Raphson's method with initial start $x_0 = 0.5$

(iv) [04] If $\vec{r}(t) = \sin 2t\ \hat{i} + \cos t\ \hat{j} - t^3\ \hat{k}$ is position vector of a moving particle at time (t), then find: a. Velocity vector $\vec{v}(t)$ b. Acceleration vector $\vec{a}(t)$ — OR (04) — Find the equation of

line passing through the point of intersection of lines x+2y=5 and x−2y=1 and parallel to line x−2y=3 and write it into slope intercept form.

(v) [04] Evaluate: ∫(2z−3)(z+2) dz — OR (04) — Find the equation of the circle passing through the points (0,4), (5,0) and (0,0)

(vi) [04] Find the solution of differential equation dy/dx = 2 + e^x y + y + 2e^x — OR (04) — Evaluate ∫ x^2 sin x dx using integration by parts method.

(vii) [04] Find the extreme values of f(x) = 1/3 x^3 + 1/2 x^2 − 6x + 5 — OR (04) — If 8x^2 − 2xy − 3y^2 = 0 is 2nd degree homogeneous equation then find: a. Pair of straight lines b. Angle between these lines

(viii) [04] Find the area of a triangle if its vertices are A(4,5), B(5,2) and C(6,7). — OR (04) — Find the area under the graph of f(x) = 6x^2 + 4x over the interval [1,3]

(ix) [04] Find the equations of tangent and normal to parabola x^2 = −8y + 8 at x = 4. — OR (04) — Find dy/dx if y = (3x+2)^(3/2) by first principle.

(x) [04] Evaluate ∫ e^θ / (e^(2θ) + 1) dθ by substitution method. — OR (04) — If x^2/16 − y^2/9 = 1 is an equation of hyperbola then find: a. Centre b. Vertices c. Foci d. Equation of asymptotes

(xi) [04] If f(x,z) = √(x^2 + z^2), then find: a. ∂f/∂x b. ∂f/∂z — OR (04) — If f(x) = 3x+2 and g(x) = 2x−3, and h(x) = f(g(x)) then find: (i) h(x) (ii) h^{-1}(x) (iii) value of x if h^{-1}(x) = 2

(xii) [04] Compute four iterations of the function f(x) = 2x^3 − 7 in the interval [0,2] by using Bisection method. — OR (04) — Find dy/dx, when 3x^3 − 6x^2y^2 − 5y^3 = 0

> Note: SECTION – C (Marks 32) — Note: Solve the following Questions. (4 x 8 = 32)

Q3. (Marks 08) Q.3 The growth of a virus is the function of time (t) (in hours) f(t) = 200e^(0.5t), then find: a. Initial number of virus at t=0 b. Number of virus after t=2 hours c. Number of virus after t=4 hours d. The time when number of virus is 200e^4. e. Is f(t) continuous at t=2 hours and t=4 hours. — OR (08) — If f(x) = 1/8 x^3 − 1/4 x^2 − 2/x^2 + 5/2, then find: a. Equation of tangent at x=2 in ax+by=c form. b. Equation of normal at x=2 c. Test the continuity of f(x) at x=0

Q4. (Marks 08) Q.4 In the given ΔABC find equation of: a. Median $C\bar{M}$ b. Altitude $C\bar{D}$ c. Right bisector of $A\bar{B}$ at mid-point M — [Graph: A(1,1), B(7,1), C(3,5)] — OR (2+2+2+1+1) — In an elliptic ground b = 1/2 a and a = 8 find: a. Vertices and co-vertices b. Foci c. Eccentricity d. Equation in standard form with centre at origin e. Draw the rough diagram of ellipse using vertices and co-vertices only. (Graph paper is not required)

Q5. (Marks 08) Q.5 Evaluate: ∫ (2x+1)/(x^3 + 3x^2 + 2x) dx — OR (08) — Solve the homogeneous differential equation 2xy dy − (3x^2 + y^2) dx = 0

Q6. (Marks 08) Q.6 Evaluate: $\int dy / (9y^3 - y)$ — OR (5+2+1) — a. Approximate $\int_0^2 (3x^2 + 2x)\, dx$ for (n=2 sub intervals) by using Simpson's Rule b. Find actual value of definite integral c. Calculate difference of both values